



(16-05-26)

End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2592666.....

Name of the Program and semester: BCA (AI/DS)/ BSc (IT)

Name of the Course: Introduction to Data Science

Course Code: TBD 201 / CAD201

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

- a. Explain the facets of Data Science. How do statistics, computer science, domain knowledge, and visualization contribute to the Data Science process? (CO1)
- b. Explain the architecture of Hadoop with a clear diagram, and describe all of its components in detail. (CO1)
- c. Explain the evolution of Data Science in detail. Discuss the different stages of its development and highlight the impact of Data Science in the modern AI-driven era. (CO1)

Q2. (2X10=20 Marks)

- a. Explain data cleaning, integration, and transformation. Discuss why these steps are essential before performing data analysis. (CO2)
- b. Explain the Data Science Process in detail. Describe each stage involved in transforming raw data into meaningful insights. (CO2)
- c. Discuss a case study on customer segmentation. Explain the steps involved from data collection to model building and interpretation of results. (CO2)

Q3. (2X10=20 Marks)

- a. A dataset is given as: 5, 8, 13, 9, 10, 12, 15, 18. Calculate the mean, mode, median, range, variance, and standard deviation of the data. Explain what these measures indicate about data dispersion. (CO3)
- b. A college aims to evaluate whether there is a significant difference in the academic performance of students enrolled in two different courses, namely BCA and MCA, in a Programming examination. For this purpose, a random sample of 15 BCA students is selected, and their average marks are found to be 78 with a standard deviation of 10. Similarly, a random sample of 15 MCA students is selected, and their average marks are found to be 72 with a standard deviation of 8. Using a significance level of 0.05 and a given critical value of $t = \pm 2.048$, test whether there is a significant difference between the mean marks of the two groups. Perform an appropriate independent t-test and clearly state the null and alternative hypotheses, calculate the degrees of freedom, determine the critical value, compute the test statistic, and provide the final conclusion. (CO3)
- c. A university claims that the average score of students in a Data Science course is 70 marks. A new teaching method is introduced to improve student performance, and a random sample of 25 students is selected. The results show that the sample mean score is 75 marks with a standard deviation of 10 marks. Using a significance level of 0.05 and a given critical value of $t = 1.711$, test whether the new teaching method has significantly increased the average score of students. Perform a right-tailed one-sample t-test by stating the null and alternative hypotheses, calculating the degrees of freedom, determining the test statistic, comparing it with the critical value, and writing the final conclusion. (CO3)

Q4. (2X10=20 Marks)

- a. Explain the concept of data visualization and discuss its importance in data science. Differentiate between exploratory and explanatory visualization with suitable examples. (CO4)



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- b. Describe various basic plotting techniques such as line charts, bar charts (vertical and horizontal), pie charts, donut charts, histograms, and box plots. Explain their use cases. (CO4)
- c. A standardized test has scores that follow a normal distribution with a mean of 50 and a standard deviation of 10. A student scores 65 marks in the test. Using the concept of z-score and standard normal distribution, determine how many standard deviations the student's score is above the mean. Also, calculate the probability that a randomly selected student scores greater than 65 marks. (z-value=0.4772). (CO3)

Q5.

(2X10=20 Marks)

- a. Explain various model evaluation metrics such as accuracy, precision, recall, and F1-score. When should each metric be preferred? (CO5)
- b. Describe the working of the following machine learning algorithms:
- Linear Regression
 - Decision Trees
 - K-Nearest Neighbors (KNN) (CO5)
- c. Explain the concept of machine learning. Differentiate between supervised and unsupervised learning with suitable real-world examples. (CO5)

Note For the question paper setters:

- Question paper should cover all the COs of the course, preferably equal distribution.
- Please specify COs against each question.